

Modeling and Verification of a
Data Aligner
using **SystemVerilog** and **UVM**

Results

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1 Introduction

The **Data Aligner** module takes in an unaligned stream of data and outputs it as an aligned stream of data based on its configuration.¹

In [this project](#), the Data Aligner is modeled and verified using the SystemVerilog HDVL and the Universal Verification Methodology (UVM).

This project is the course project of this Udemmy course: [Design Verification with SystemVerilog/UVM](#).

All tests were run many times with random seeds.²

This document contains:

1. Testing results.
2. Coverage results.
3. Assertions (checking) results.

¹ For more information about the Data Aligner module (its interfaces, registers, functionality, and more), see its specification: *Aligner Datasheet - v1.1*.

² See: *Data Aligner - Architecture of The Verification Environment* and *Data Aligner – Verification Plan*.

2 Testing Results

In general, the DUT passed all tests successfully, *but* it failed twice; two bugs were discovered.

2.1 Bug #1

After running simulation with this test: `align_test_reg_access` and this seed: 1 (default), the following error and infos were reported:

```
UVM_ERROR [...] @ 125: reporter [RegModel] Register "reg_block.IRQEN" value read from DUT (0x000000000000001f) does not match mirrored value (0x0000000000000000)

UVM_INFO [...] @ 125: reporter [RegModel] Field RX_FIFO_EMPTY (reg_block.IRQEN[0:0]) mismatch read=1'h1 mirrored=1'h0
UVM_INFO [...] @ 125: reporter [RegModel] Field RX_FIFO_FULL (reg_block.IRQEN[1:1]) mismatch read=1'h1 mirrored=1'h0
UVM_INFO [...] @ 125: reporter [RegModel] Field TX_FIFO_EMPTY (reg_block.IRQEN[2:2]) mismatch read=1'h1 mirrored=1'h0
UVM_INFO [...] @ 125: reporter [RegModel] Field TX_FIFO_FULL (reg_block.IRQEN[3:3]) mismatch read=1'h1 mirrored=1'h0
UVM_INFO [...] @ 125: reporter [RegModel] Field MAX_DROP (reg_block.IRQEN[4:4]) mismatch read=1'h1 mirrored=1'h0
```

After debugging and analysis, I found that the reason of this error is:

In *Aligner Datasheet - v1.1* (and my model), the reset (initial) value of the IRQEN register is 0, but in the RTL implementation, the reset (initial) value of the IRQEN register is 32'h00_00_00_1F.

The RTL can be fixed by following modification:

In *cfs_regs.v*, replace the following ones with zeros.

```
/* line 175 */      irqen_rx_fifo_empty <= 1;
                   irqen_rx_fifo_full  <= 1;
                   irqen_tx_fifo_empty <= 1;
                   irqen_tx_fifo_full  <= 1;
                   irqen_max_drop      <= 1;
```

2.2 Bug #2

After running simulation with this test: `align_test_illegal_rx` and this seed: 1 (default), the following error was reported:

```
UVM_ERROR align_scoreboard.sv(186) @ 545505: uvm_test_top.m_algn_env.m_algn_scoreboard [m_algn_scoreboard]
IRQ_information MISMATCH:: num_ones_in_expected_irq_fifo: 1 - num_ones_in_actual_irq_fifo: 0

UVM_ERROR align_scoreboard.sv(198) @ 545505: uvm_test_top.m_algn_env.m_algn_scoreboard [SCOREBOARD RESULTS]
Mismatches:: ERR; number of mismatches: 1
```

After debugging and analysis, I found that the reason of this error is:

In *Aligner Datasheet - v1.1* (and my model), if `IRQ.MAX_DROP` and `IRQEN.MAX_DROP` equal 1 then `irq` shall be set, but in the RTL implementation, this condition is not imposed.

The RTL can be fixed by following modification:

In *cfs_regs.v*, add the following code (in **bold green**):

```
/* line 164 */    assign irq =

                (edge_rx_fifo_empty & irqen_rx_fifo_empty) |
                (edge_rx_fifo_full  & irqen_rx_fifo_full)  |
                (edge_tx_fifo_empty & irqen_tx_fifo_empty) |
                (edge_tx_fifo_full  & irqen_tx_fifo_full)  |
                (edge_max_drop   & irqen_max_drop);
```

3 Coverage Results

[Table 1](#) shows coverage results.

Table 1 Coverage results

#	Coverage Collector	covergroup	Coverpoint/ cross	Coverage (%)	Test and Seed ³
1	m_apb_coverage_collector	cover_reg	CTRL_OFFSET	100.00%	algn_test_reg_access (-1993361420)
			CTRL_SIZE	100.00%	
			CTRL_OFFSET_x_CTRL_SIZE	100.00%	
			IRQ_MAX_DROP	100.00%	algn_test_illegal_rx (1)
2	m_md_coverage_collector_master	cover_item	offset	100.00%	algn_test_random (1)
			size	100.00%	
			offset_x_size	100.00%	
			response	100.00%	

³ Tests and seeds used to obtain these results. The seed is enclosed in round brackets.

4 Assertions Results

The DUT succeeded in all assertions. No errors were reported.